

AALYA KRISHNAN U

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Full Stack Software Developer with expertise in Python, Flask, React.js and real-time data pipelines. Hands-on experience across the complete Software Development Life Cycle (SDLC) through 2 internships and 9+ projects spanning machine learning, backend development and data engineering.

EDUCATION

Atria Institute of Technology — B.E. in Computer Science & Engineering (VTU)	CGPA: 8.33/10 2025
New Horizon PU College (PCMB)	86.3% 2021
Oxford English High School	89.92% 2019

INTERSHIP EXPERIENCE

ATANE — Software Developer	Dec 2024 – Apr 2025
- Trained and tested a CNN model on the GTSRB dataset (50,000+ images, 43 classes) using OpenCV and TensorFlow, achieving ~97% detection accuracy for real-time traffic sign recognition - Optimized existing Python codebase through hyperparameter tuning and preprocessing improvements, reducing model inference time for real-time detection performance	
Varcons Technologies Pvt. Ltd. — Full Stack Web Developer	Oct 2023 – Dec 2023
- Designed and implemented UI for a 3-level interactive web-based escape room game across 3-4 screens using HTML, CSS and Figma, following SDLC from design to deployment - Contributed to front-end development and visual design implementation as part of the development team	

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, PHP, HTML, CSS
Backend Development: Flask, Django, REST APIs, API Integration, NLTK
Frontend Development: React.js, Firebase SDK, HTML, CSS, JavaScript
Data Engineering: Apache Kafka, Pandas, NumPy, SQLite, MySQL, ETL Pipelines
Machine Learning: TensorFlow, PyTorch, Keras, OpenCV, CNN, Deep Learning, Data Augmentation
Tools & Practices: Git, GitHub, Postman, Figma, VS Code, Jupyter Notebook, Power BI, SDLC, Debugging, OOP

PROJECTS

QUICKBITE — Real-Time Food Delivery Pipeline 	2026
- Built a real-time food delivery system using Apache Kafka for message streaming across customer, restaurant and admin panels, implementing producer-consumer architecture in Python for asynchronous order processing - Developed 3 Flask web interfaces with REST API backend, SQLite database and payment flow supporting Card, UPI and Cash on Delivery with OTP verification simulation	
TASKMASTER — Task Management Web App 	2026
- Built a React.js task management application with Firebase Authentication supporting login, signup and password reset, and Firestore real-time database enabling live task creation, editing, deletion and deadline tracking - Implemented real-time online user presence tracking using Firestore onSnapshot listeners with automatic session cleanup on logout	
GAME FOR ADHD KIDS — Final Year Project	2025
Designed and developed a Unity-based gamified platform for children with ADHD comprising 3 interactive mini-games with progressive difficulty levels (easy, medium, hard) to enhance focus, engagement and cognitive development	
ROAD SIGN DETECTION & IDENTIFICATION USING CNN	2025
Developed a real-time traffic sign detection system training a CNN on the GTSRB dataset (50,000+ images, 43 classes) using OpenCV achieving ~97% accuracy through data preprocessing and parameter optimization	
STUDENT MANAGEMENT USING REST API 	2025
Developed a RESTful API using Flask and SQLite implementing 5 endpoints for complete CRUD operations — Create, Read, Update, Delete — on student records with JSON request/response handling and HTTP status compliance	

TELEMEDIX — Virtual Clinic Platform with Integrated Chatbot

2024

- Built a virtual clinic platform with an ML-powered chatbot using NLTK and Keras, implementing NLP techniques including tokenization, lemmatization and bag-of-words for intent classification with 25%+ confidence threshold filtering
- Developed Flask backend API serving chatbot responses for appointment booking, medical queries and specialist contact directory

REAL-TIME FACE EMOTION DETECTION

2024

- Developed a real-time facial emotion detection system training a 3-layer CNN on the FER2013 dataset (35,000+ images) to classify 7 emotion categories using TensorFlow and OpenCV with data augmentation for improved generalization
- Built a Tkinter desktop application with live webcam feed processing at ~30ms per frame using Haar Cascade face detection pipeline

ONLINE BOOK SHOPPING WEBSITE

2023

- Developed a SQL-based online bookstore database system implementing user authentication, inventory management and order processing modules with normalized relational schema and CRUD operations

IoT-BASED SMART PARKING SYSTEM

2022

- Developed an IoT-enabled smart parking system in C using motion detection sensors to monitor and track availability of 4 parking slots in real time

CERTIFICATIONS

- Data Science for Beginners — NASSCOM
- Google Cybersecurity — Coursera
- Full Stack Web Development — Varcons Technologies
- Cloud Computing — NPTEL